Dongho Kim

Email Website LinkedIn

Research Interests

Reliable AI, Sociotechnical Evaluation of AI Systems, Human–AI Collaboration, Computational Social Science

Education

Northwestern University — **Evanston, IL**
M.S. in Statistics and Data Science — *Sep 2025 – Expected Jun 2027*

Dongguk University — **Seoul, South Korea**
B.B.A. in Business Administration; B.S. in Data Science Software — *Mar 2019 – Feb 2025*

University of California, San Diego — **San Diego, CA**
Exchange Student, Business Economics and Data Science — *Sep 2022 – Jun 2023*

Publications and Preprints (* equal contribution)

- Ham, M.*, **Kim, D.***, Lee, C.*, Kim, M. J., Zhang, Y., Wang, J., and Liu, H. *SPINE: Bridging the Cyber-Physical Gap with Agentic AI.* arXiv:2607.13049, 2026. Under review. Project page.
- Li, W.*, Chen, H.-Y.*, Rehemtulla, N.*, Shah, V., **Kim, D.**, Wu, D., Lin, Q., Miller, A., and Liu, H. *StarEmbed: Benchmarking Time Series Foundation Models on Astronomical Observations of Variable Stars.* ICML 2026 (poster); oral presentation and Outstanding Paper Award, AI4Physics Workshop at ICML 2026; arXiv:2510.06200.

Research Experience

Graduate Research Intern — **Sep 2025 – Present**
MAGICS Lab, Northwestern University; Advisor: Prof. Han Liu — *Evanston, IL*

- Co-developed SPINE (co-first author), an agentic harness that walks robotics novices through diagnosing and repairing real robots, with step-by-step checks until live validation passes.
- Designed 12 fault scenarios across two robot platforms; novice-operated SPINE resolved all implanted faults and cut mean time to teleoperation by 30–38% versus experts using Claude Code.

Research Collaborator — **Oct 2025 – Present**
NSF–Simons SkAI Institute, Northwestern University — *Evanston, IL*

- StarEmbed: benchmarked zero-shot embeddings from time-series foundation models (TSFMs) on ∼40,000 expert-labeled variable-star light curves for classification and clustering.
- Pretrained a Transformer-based TSFM on irregularity-augmented data and explored timestamp-based rotary position embeddings and Mamba to address its limits on irregular sampling.
- Developing Star-Bench to evaluate AI agents on variable-star analysis tasks using multi-survey observations.

Research Assistant — **Dec 2023 – Aug 2024**
Intelligence and Optimization in Networks Lab; Advisor: Prof. Hyeryung Jang — *Seoul, South Korea*

- Implemented a hybrid RNN–FinBERT pipeline for stock-return prediction across 88 firms, combining price histories with text-derived sentiment features.

Research Projects

Sci2Pol-Agent: Agentic Workflows for Science-to-Policy Writing — **Mar 2026 – Present**
Originated in IEMS 447 (Prof. Dashun Wang) — *Evanston, IL*

- Prototyped training-free, multi-role agent workflows that route science-to-policy writing through evidence extraction, claim verification, and revision, and compared them with direct prompting on Sci2Pol-Bench.
- Developing and evaluating agentic workflows for science-to-policy writing, focusing on source fidelity and the quality of the resulting briefs.

Information Contagion and Market Connectivity — **Sep 2025 – Dec 2025**
COMP SCI 496: Social Network Analysis, Prof. Noshir Contractor — *Evanston, IL*

- Constructed a news co-mention network of 117 NASDAQ technology firms (2019–2020) and estimated volatility contagion with panel regressions and a stochastic actor-oriented model (RSiena).

Presentations

Frontier Agents for Variable-Star Analysis and Classification **Sep 2026**
Poster presentation, Open SkAI 2026; SkAI Institute *Chicago, IL*

Technical Skills

Programming: Python, PyTorch, R, SQL, Git, LaTeX

Machine Learning: Model pretraining and fine-tuning, LLM agent development, benchmark design and ablations

Statistics and Data: Regression, hypothesis testing, feature engineering, experimental design, network analysis

Professional and Teaching Experience

Teaching Assistant **Sep 2023 – Jun 2024**
Department of Business Administration, Dongguk University; Prof. Sanghyeon Sung *Seoul, South Korea*

- Courses: Organizational Behavior, Human Resource Management, and Diversity Management.

Operations Assistant **Nov 2021 – Mar 2022**
M&E Parts *Seoul, South Korea*

- Built an inventory analytics system to monitor stock levels, identify excess inventory, and support supply planning.

Sergeant, Military Service **Jan 2020 – Sep 2021**
Republic of Korea Ministry of National Defense *South Korea*

- Led a team managing mission-critical communication equipment and standardized operational-readiness reporting.

Awards and Scholarships

- Outstanding Paper Award, AI4Physics Workshop at ICML 2026 (oral presentation), 2026 – *StarEmbed*
- Excellence Award, Seoul City AI Video Content Competition, 2024
- Top 3%, LG AI Research B2B Sales Opportunity Prediction Competition, 2024
- Exchange Program Scholarship, Dongguk University, 2023
- Academic Excellence Scholarship, Dongguk University, 2022

Service and Languages

- International Student Mentor, International Buddy Program, Dongguk University, 2019
- Korean: native; English: proficient (GRE Q 170, V 158; IELTS 7.0); Chinese: intermediate (HSK 4)